S. Jaya Krishna Teja

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Professional Summary

Lead Mechanical Design Engineer with 4.5 years of manufacturing-focused R&D experience in Air compressors, Air treatment units, Combustion chambers, gas turbines, and electro-mechanical systems. Demonstrated expertise in designing robust, manufacturable products using SolidWorks with strong proficiency in DFM, DFA, DFMEA, and tolerance analysis. Proven track record of delivering high-performance products, mechanical components to Railways OEMS and IR from concept through production while collaborating effectively with vendors, quality teams, and cross-functional stakeholders. Seeking to leverage technical design capabilities and manufacturing knowledge as a Lead Mechanical Engineer at Hitachi Energy.

Technical Skills

CAD & Design Tools: SolidWorks| AutoCAD | CATIA | Space Claim
Design & Documentation: GD&T (ASME Y14.5)| Tolerance Stack-Up Analysis | 3D & 2D Detailing | MBD/PMI | DFM | DFA | DFMEA |FMECA | BOM Generation | BOM Check | ROHS compliance.
Manufacturing Expertise: Sheet Metal Fabrication | Welding Processes | Assembly Operations |AL castings
Testing & Validation: Prototyping |TYPE TEST | First Article Inspection (FAI) | Vibration Analysis(IEC61373)
Quality & Compliance: ISO 9001| EN15085-2 -CPD| ISO 13092 | 6σ Green Belt | PPAP | PFMEA | 7QC Tools
PLM & Software: Enovia (3D Experience)|MATLAB |Python |C++ | Microsoft Project| Excel |PowerPoint
Project Management: Project Planning | Vendor Management |Stakeholder Communication | Budget Allocation | Risk Management| Cost Optimization | RACI |

Professional Experience

Lead Mechanical Design Engineer – NPD
ELGi Equipments Ltd. | Coimbatore, India Aug 2025 – Present
Railways Product Integration – Air Compressors & Air Dryer Systems

- Delivered 3 air compressor and treatment unit products for railway applications through full design and development ownership
- Achieved 100% customer satisfaction across 3 projects by maintaining strict adherence to EN15085 welding standards and meeting all budget and performance objectives.
- Designed complex mechanical assemblies including sheet metal enclosures, machined precision components, and plastic housings using SolidWorks ensuring manufacturability & cost-effectiveness.
- Applied DFM, DFA, and tolerance stack-up analysis principles to optimize component designs, reducing manufacturing complexity and assembly time by 17%.
- Managed vendor collaboration for electro mechanical component development and valve component manufacturing, coordinating technical specifications and quality with suppliers to ensure
- Conducted First Article Inspections (FAI) and validation testing, improving customer satisfaction by 13% through rigorous quality verification and proactive issue resolution.
- Executed risk assessments and implemented mitigation strategies, achieving 90% improvement in supply chain reliability and reducing quality-related delays.
- Drove continuous improvement initiatives resulting in 15% cost optimization through value engineering and supplier rationalization.

Sr. Mechanical Design Engineer

ELGi Equipments Ltd. | Coimbatore, India Aug 2022 – Aug 2025

Air Compressor Systems Development (Recip)

- Designed and developed electro-mechanical components including castings, machined parts, plastic moulded components, and sheet metal assemblies for air compressor systems using SolidWorks
- Delivered 300+ detailed engineering drawings with comprehensive GD&T annotations per ASME Y14.5 standards, ensuring manufacturing accuracy and assembly compatibility.
- Managed BOM creation and Engineering Change Notices (ECN) through PLM system, improving design traceability and documentation accuracy by 18%
- Conducted DFMEA sessions with cross-functional teams to identify and mitigate potential failure modes, enhancing product reliability.
- Participated and conducted prototyping activities and mechanical testing to validate design performance under operational conditions

Freelance & Additional Projects

Freelance Design Mentor | AMRITA University Bengaluru Sept 2025 – Present

Mentoring mechanical design and development of ATV vehicle for BAJA 2025 competition, guiding students in applying DFM/DFA principles

Freelance Medical Device Consultant | MIT, US Apr 2025 - Jul 2025

Developed catheter design concepts and conducted blood flow validation analysis in collaboration with research scholar

Research Consultant – Combustion Systems

Technical University of Eindhoven | Netherlands Feb 2022 - Jul 2022

- Applied computational fluid dynamics (CFD) and thermal analysis techniques for gas turbine component design optimization.
- Collaborated with international research teams to develop innovative mechanical solutions for high-temperature engine components
- Analysed thermo-mechanical fatigue (TMF) effects on materials under various operational conditions, supporting robust design decisions.
- Utilized data-driven approaches including regression analysis to optimize combustion system performance

Research Associate – Micro Gas Turbines

Turbine Electric Company GmbH & Hochschule Darmstadt | Germany May 2021 - Mar 2022

- Designed critical rotating mechanical components for multi-fuel compatible micro gas turbine combustion chambers
- Created detailed 3D models using CATIA V5 and verified designs during prototype development and manufacturing feasibility assessments
- Conducted comprehensive cold and hot flow simulations using ANSYS FLUENT to predict component performance and validate design integrity
- Collaborated with regional suppliers for prototype manufacturing and component development, ensuring design intent was maintained through production
- Managed project timelines and technical deliverables while coordinating with global engineering teams

Design & CFD Intern

Larsen & Toubro Defence | Coimbatore, India Sept 2020 – Dec 2020

- Performed design calculations and CFD analysis for rocket propulsion mechanical components

- Applied GD&T principles to 3D SolidWorks models to ensure manufacturing accuracy and assembly compatibility
- Supported cross-functional engineering teams with technical documentation and design verification activities

Certifications & Training

- EN15085-2 – Railway Design Welding Standard – TUV NORD
- GD&T and Tolerance Analysis – ELGi Learning Management System
- Six Sigma Green Belt Certified
- Project Management Fundamentals – LinkedIn Learning and Google- Coursera
- Lean Principles & Continuous Improvement – LinkedIn Learning

Achievements

- Achieved IRIS Silver certification consecutively for 2022-2025
- Received "Best Performer Award" at ELGi for successful completion of critical customer project ahead of schedule
- Delivered 95% of projects within timeline and budget constraints through effective planning and execution.

Education

Darmstadt University of Applied Sciences | Germany Mar 2021 - Mar 2022

Exchange Program - Master Thesis: New Product Development (Micro Gas Turbine)

GPA: 4.0/4.0

Amrita Vishwa Vidyapeetham | India Aug 2019 - Feb 2021

Master of Technology in Engineering Design

Distinction with Scholarship | CGPA: 8.5/10.0

Amrita Vishwa Vidyapeetham | India Jun 2014 - May 2018

Bachelor of Technology in Mechanical Engineering

CGPA: 7.0/10.0